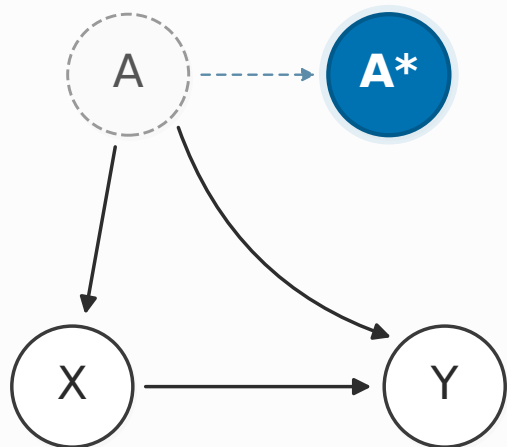


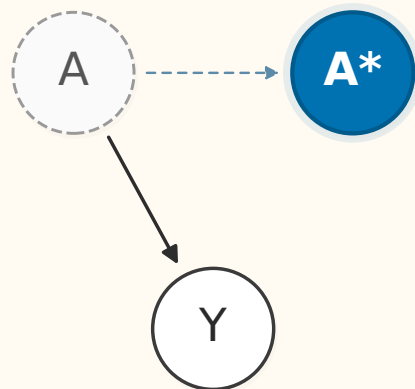
(a) Confounding

Attenuation (toward OVB)



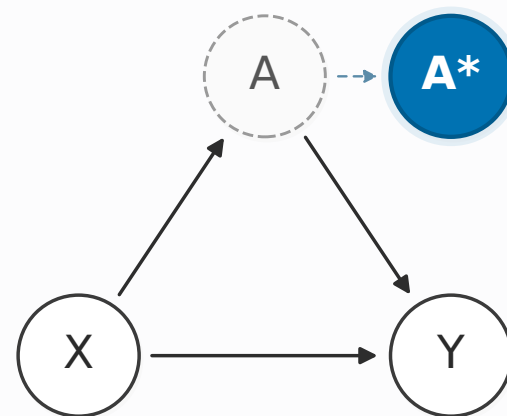
(b) Exposure

Sign reversal (83.4%)



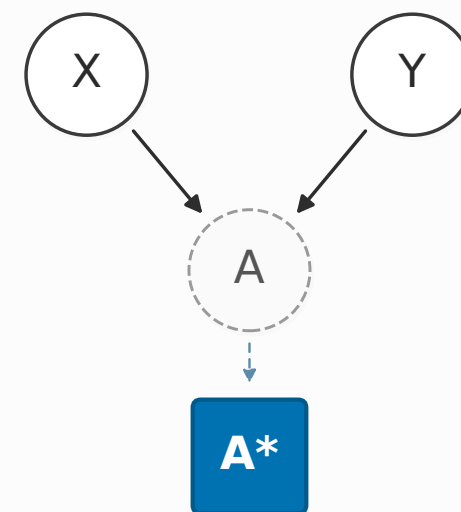
(c) Mediation

Amplification



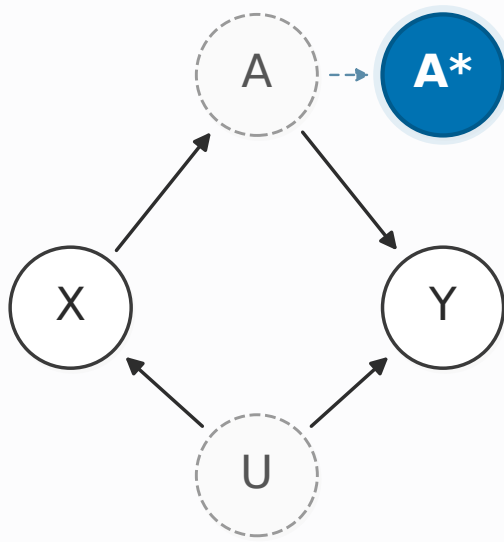
(d) Collider

Attenuation (no sign flip)



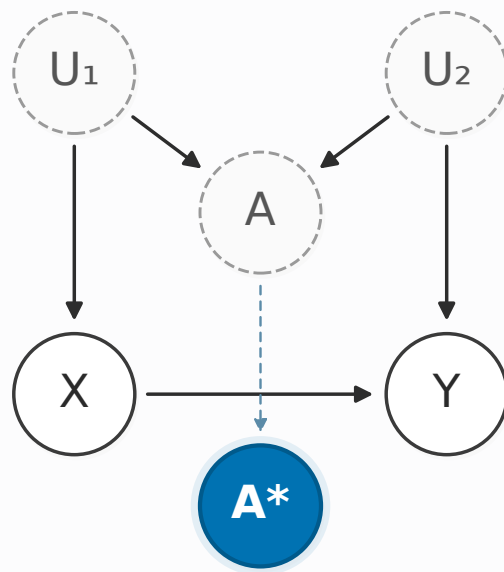
(e) Front-door

Moderate sensitivity



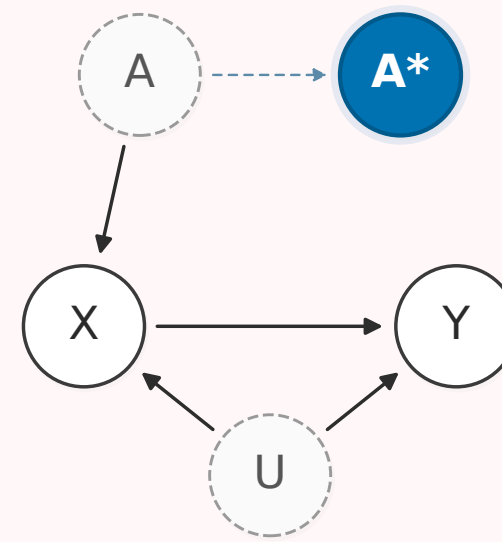
(f) M-bias

Immune



(g) IV (Instrument)

Fragile (Wald amplification)



○ Observed ○ Latent ● Misclassified (A^*) ■ Conditioned on (A^*) → Causal - - - Measurement